

On declarative and polar complements in Persian

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Abstract

In Persian, anti-rogative predicates can select for an embedded clause with the polarity particle *aya* in certain contexts. We characterize these contexts as non-veridical, and we propose that the particle does not introduce a polar question: the *ke aya*-complement denotes a unique proposition and conveys that the latter is positively or negatively decided in the attitude state of the selecting anti-rogative predicate. In veridical contexts, the *ke aya*-complement is ruled out by the competition of the corresponding declarative complement, which conveys positive decidedness and thus determines strictly more informative truth conditions. In non-veridical contexts, however, the informativeness competition is suspended and *ke aya*-complements are allowed. We argue that the polar complement interpretation constitutes a weakening of the interpretation of *aya* as the polar question operator.

Keywords: anti-rogative predicates, interrogative particles, question embedding, semantic selection

1 Introduction

Clause-embedding predicates are traditionally classified into rogative, anti-rogative, and responsive. As cited in Spector and Egré (2015), according to Lahiri (2002) rogative predicates are those which take interrogative complements but not declarative ones. Verbs such as *wonder*, *ask* and *investigate* are among the rogatives. The following example illustrates the use of an interrogative complement to rogative predicate in English:

- (1) Bill wonders whether / *that Carl left.

Anti-rogatives are those predicates that can take only declarative clauses as their complements in embedding contexts. In the following example, *believe* as an anti-rogative predicate illustrates this feature:

- (2) Bill believes that /*whether Carl left.

Rogative and anti-rogative predicates show a restricted selection: the former select for type $\langle st, t \rangle$ questions, the latter select for type $\langle s, t \rangle$ propositions. On the other hand, responsive predicates allow for both types, which makes them apparently ambiguous: both declarative and interrogative complements can be used, and in the second case, the predicate expresses a relation between an attitude holder and an answer to the embedded question (Karttunen 1977; Lahiri 2002; Spector and Egré 2015; Uekagi 2015, 2019; Mayr 2017; Uegaki and Sudo 2019; Theiler, Roelofsen and Aloni 2019; Roelofsen and Uegaki 2021).¹

The two-fold selection of responsive predicates is particularly interesting when we consider the distribution of the interrogative particle *aya* in Persian.

Aya appears optionally in main polar questions demanding an affirmative or a negative answer; when present, it adds emphasis to the question. *Aya* appears in different positions, due to its free position possibility:²

- (3) a. Aya Ali midone (ke) dustash raftand?
Pm Ali know-3S (that) friends-poss went-pl
- b. Ali aya midone (ke) dustash raftand?
Ali PM know-3S (that) friends-poss went-pl
- c. Ali midone aya (ke) dustash raftand?
Ali know-3S Pol.M (that) friends-poss went-pl?
'Does Ali know that his friends left?'
- d. Ali midone ke aya dustash raftand?
Ali know-3S that Pol.M friends-poss went-pl
'Does Ali know whether his friends left?'

In examples (3a-c), where *aya* occurs in the matrix question, the embedded declarative clause is introduced by the optional complementizer morpheme *ke*, which is equivalent to English *that*. In (3d), instead, *aya* appears in the embedded interrogative clause, and in this case the complementizer *ke* is obligatory and precedes *aya*.

In this paper we investigate the distribution of (ke)-aya complements with different types of attitude verbs. We show that anti-rogative predicates like *fekr kardan* (to think) can select *ke aya*- clauses in certain non-veridical environments, thus constituting an apparent exception to the expected pattern (§2). We propose that in these contexts, *ke aya*- clauses are not embedded polar questions, but rather polar complements, which introduce a single proposition just like declarative complements. We then provide an analysis of the distribution of *ke*- and *ke aya*-complements in terms of their relative informativeness (§3).

¹Responsive predicates are of two types: veridical (such as *know*) and non-veridical (such as *tell*). In the case of veridical responsives, the predicate expresses a relation between an attitude holder and the true answer, while for non-veridicals this relation is with a potential answer.

²We gloss *aya* as a polarity marker (PM).

In §4 we consider rogative predicates, and we propose that *ke aya*-clauses are actually ambiguous between the polar complement meaning and a proper polar question. In §5 we discuss some potential extensions of our approach, while in §6 we discuss an alternative analysis based on the assumption that *ke aya*-clauses always denote polar questions. Finally, in §7 we offer some concluding remarks.

2 Empirical data

2.1 The distribution of *aya*

As the examples in (3) above show, in Persian the responsive predicate *danestan* (to know) selects both complements with the optional *ke* complementizer and *ke-aya*-complements. On the other hand, the rogative verb *porsidan* (to ask) selects for *ke-aya*-clauses (4), but it cannot select *ke*-clauses, either in a main declarative (5a) or in a main question (5b). (The reason for this observation will become clear in a moment.)

- (4) Ali az man porsid ke aya dustash raftand.
 Ali from me asked -3S that PM friends-poss went-pl
 'Ali asked me whether his friends left.'
- (5) a. * Ali az man porsid ke dustash raftand.
 Ali from me asked -3S that friends-poss went-pl
 b. * Ali porsid ke dustash raftand?
 Ali asked-3S that friends-poss went-pl

The natural hypothesis is that *ke*-clauses denote a proposition and *ke aya*-clauses denote a polar question: under this view, *aya* corresponds to the polar question operator, which turns the prejacent proposition *p* into the polar set $\{pp, \neg p\}$.

Persian anti-rogative predicates such as *fekr kardan* (to think), *bavar dash-tan*, *bavar kardan* (to believe), *ghabul dashtan* (to agree) select for *ke*-clauses only in main declaratives, as expected:

- (6) Sara fekr mikonad / bavar darad ke dustash raftand.
 Sara think do-3S/ believe-3S that friends-poss went-pl
 'Sara thinks/believes that her friends left.'
- (7) * Sara fekr mikonad / bavar darad ke aya dustash raftand.
 Sara think-3S / believe-3S that PM friends-poss went-pl

However, there are exceptions to the expected pattern with anti-rogative predicates. As the following examples show, *ke-aya*-complements are possible in main interrogative clauses:

- (8) Sara fekr mikonad ke aya dustash raftand?
 Sara think-3S that PM friends-poss went-pl
 lit. 'Does Sara think whether her friends left?'

In addition, *ke-aya*-complements are possible when the main clause is negated:³

- (9) Sara fekr nemikone / bavar nadarad ke aya dustash raftand.
 Sara think not-does/ believe not that Pol.M friends- poss went-pl
 lit. 'Sara does not think/believe whether her friends left.'

Finally, *ke aya*-clauses are possible in the scope of epistemic modals:

- (10) Sara bayad fekr konad ke aya Ali rafte ast.
 Sara probably think do-3S that Pol.M Ali gone is
 lit. 'Sara probably thinks whether Ali has left.'
- (11) Sara shayad fekr konad ke aya Ali rafte ast.
 Sara perhaps think do-3S that Pol.M Ali gone is
 lit. 'Sara might think whether Ali has left.'

2.2 A generalization

The data in (4)-(11) lead us to the following generalizations.

On the one hand, *ke*-complements (where the particle is optional) show the distribution expected for declarative clauses: they can be selected by responsive and anti-rogative predicates, but not by rogative ones. On the other hand, the classification of *ke-aya*-complements is less straightforward. When the main clause is declarative, they can only be licensed by responsive and rogative predicates, which is the expected distribution of interrogative clauses. However, a *ke-aya*- complement can be licensed by an anti-rogative predicate if the matrix clause contains one of the following operators:

- i) the polar question operator, as in (8);
- ii) matrix clause negation, as in (9);
- iii) epistemic modals, as in (10)-(11).

All of these operators qualify as non-veridical in the sense of Giannakidou (1998):

- (12) An operator O is non-veridical if and only if O applied to a proposition p does not entail p: $\neg [O(p) \subseteq p]$

We can therefore formulate a second generalization:

- (13) In Persian, anti-rogative predicates can license a *ke-aya*-complement only if they are in the scope of a non-veridical operator.

³It must be mentioned that with *bavar dashtan* / *kardan*, the negative coda *ya na* is required:

- (1) Sara bavar mikonad ke aya dustash raftand ?*(ya na)?
 Sara believe-3S that PM friends-poss went-pl or not
 'Does Sara believe whether her friends left or not?'

With *fekr khardan*, instead, *ya na* is optional. We have no account for this observation.

3 Declarative and polar complements

The generalization in (13) seems to challenge the well-established distinction between anti-rogative and responsive predicates. In this section, we will argue that the counterexample is only apparent: *ke-aya*-clauses embedded under anti-rogative predicates do not denote polar questions, but rather, polar complements, a novel type of complement clause that will be defined below. Emphasizing the role of the embedded complementizer, we propose the following hypotheses:

1. Declarative *ke*-complements convey that the embedded proposition is positively decided in the attitude state introduced by the matrix verb;
2. polar *ke aya*-complements, instead, convey that the embedded proposition is decided *either positively or negatively* in the relevant attitude state.

Based on this hypothesis, we reduce the ungrammaticality of *ke aya*-complements in veridical contexts to the competition of the corresponding declarative *ke*-complement, which is strictly more informative.

3.1 Decidedness

To develop our proposal, we define the following relations between a belief state (a set of worlds) and a sentence *S*. We use the abbreviation $BEL(y,w)$ for the belief state of an individual *y* in world *w*.

Consider now a sentence *S* denoting a proposition *p* and a belief state $BEL(y,w)$. We define decidedness as follows (adapting from Farkas 2003):

- (14) Decidedness
- a. *p* is positively decided in $BEL(y,w)$ if and only if all the worlds in $BEL(y,w)$ are *p*-worlds, i.e., $BEL(y,w) \subseteq p$
 - b. *p* is negatively decided in $BEL(y,w)$ if and only if no world in $BEL(y,w)$ is a *p*-world. i.e., $BEL(y,w) \cap p = \emptyset$
 - c. *p* is decided in $BEL(y,w)$ if and only if *p* is either positively or negatively decided in $BEL(y,w)$, i.e. $BEL(y,w)$ contains only *p*-worlds or only non-*p* worlds. Otherwise, *p* is undecided in $BEL(y,w)$, i.e., $BEL(y,w)$ contains both *p*- and $\neg p$ worlds.

In natural languages, the belief relation is expressed through a propositional attitude predicate that takes as arguments a noun phrase – denoting the attitude holder – and a declarative complement clause. The standard analysis of anti-rogative predicates like *believe*, adapted to our terminology, states that the unique alternative *p* denoted by the complement clause is positively decided in the subject’s attitude state at the relevant evaluation world (cf. Farkas 2003; Anand and Hacquard 2012):

$$(15) \quad \llbracket believe \rrbracket^w = \lambda p. \lambda y. BEL(y, w) \subseteq p.$$

Under this definition, *believe* (as well as other anti-rogative predicates) cannot select for a set of propositions, corresponding to the denotation of an interrogative complement clause.

3.2 Our proposal

Going back to Persian complement clauses, we assumed above that *ke-aya*-complements are embedded polar questions denoting the set $\{p, \neg p\}$. But then, the conclusion that we just drew about anti-rogative predicates is *prima facie* incompatible with the observation that such predicates can select for a *ke-aya*-clause in non-veridical contexts, as exemplified in (8), repeated here as (16):

- (16) Sara fekr mikonad ke aya dustash raftand?
 Sara think-3S that PM friends-poss went-pl
 lit. 'Does Sara think whether her friends left?'

This empirical observation can be reconciled with the above characterization of anti-rogative predicates only if *ke-aya*-clauses denote a single proposition, rather than a polar set: in other terms, the presence of the particle *aya* does not imply the presence of the polar question operator in the complement clause. We dub *ke-aya*-clauses *polar complements*, to distinguish them from embedded questions.

What is then the semantic contribution of *aya*, or more precisely, of *ke-aya*?⁴ We propose that it marks the status of the complement clause with respect to the attitude state introduced by the matrix predicate: in particular, it conveys that the unique alternative in the denotation of the complement clause is decided – either positively or negatively – in that attitude state.

To achieve this, the denotation of the attitude verb in (15) must be revised: the relation between the attitude state and the propositional content of the complement clause cannot be encoded directly in the lexical semantics of the matrix predicate. We therefore propose that the matrix predicate only contributes the subject's relevant attitude state, whose relation to a given propositional content is specified by the complement clause.⁵

We propose that declarative *ke*-complements take as input an attitude state q (a set of worlds) and convey that the proposition denoted by the sentence radical S – call it p – is positively decided in that attitude state:

$$(17) \quad \llbracket ke-S \rrbracket = \lambda q. q \subseteq p$$

The matrix anti-rogative verb selects for a property P of attitude states (type $\langle st, t \rangle$) and conveys that P holds of the subject's attitude state at the evaluation world w , e.g.:

⁴Recall that *ke* obligatorily precedes *aya* in complement clauses, but without *aya* it is optional.

⁵This proposal is in the spirit of Kratzer (2006), Moulton (2009).

$$(18) \quad \llbracket fekr - kardan \rrbracket^w = \lambda P. \lambda y. P(\text{BEL}(y, w))$$

The composition is straightforward:

$$(19) \quad \llbracket fekr - kardan \rrbracket^w(\llbracket ke - S \rrbracket) = \lambda P. \lambda y. P(\text{BEL}(y, w)) (\lambda q. q \subseteq p) = \lambda y. \text{BEL}(y, w) \subseteq p$$

As a result, the composition of *fekr kardan* and a declarative *ke*-complement conveys that all the worlds in the belief state of the matrix subject are *p*-worlds, ultimately yielding the same truth conditions as the standard Hintikka-style analysis.

We propose that the combination of particles *ke aya* conveys that the complement proposition *p* is decided – either positively or negatively – in the attitude state provided by the matrix verb: this means that either all the worlds in that state are *p*-worlds, or else no world is a *p*-world (cf. 14c):

$$(20) \quad \llbracket ke - aya - S \rrbracket = \lambda q. q \subseteq p \vee q \cap p = \emptyset$$

$$(21) \quad \llbracket fekr - kardan \rrbracket^w(\llbracket ke - aya - S \rrbracket) = \lambda P. \lambda y. P(\text{BEL}(y, w)) (\lambda q. q \subseteq p \vee q \cap p = \emptyset) = \lambda y. \text{BEL}(y, w) \subseteq p \vee \text{BEL}(y, w) \cap p = \emptyset$$

3.2.1 Anti-rogatives in veridical contexts

Consider now the veridical context of a main declarative clause in (6) and (7), repeated here as (22) and (23):

$$(22) \quad \begin{array}{l} \text{Sara fekr mikonad / bavar darad ke dustash raftand.} \\ \text{Sara think do-3S/ believe-3S that friends-poss went-pl} \\ \text{'Sara thinks/believes that her friends left.'} \end{array}$$

$$(23) \quad \begin{array}{l} * \text{Sara fekr mikonad / bavar darad ke aya dustash raftand.} \\ \text{Sara think-3S / believe-3S that Pol.M friends-poss went-pl} \end{array}$$

A crucial consequence of the above analysis is that the proposition obtained by combining the anti-rogative predicate with a declarative *ke*-complement, as in (22), is strictly more informative than in the case of a *ke-aya*- complement (23): for any evaluation world *w*, if the complement proposition *p* is positively decided in $\text{BEL}(S, w)$ it is also decided in $\text{BEL}(S, w)$, though not viceversa.

We therefore propose that (23), featuring a polar complement, is not ruled out as syntactically or semantically ill-formed, but is rather ruled out by the competition with the more informative sentence (22) featuring a declarative complement. According to the general principle of maximizing informativeness, ultimately related to Grice's Maxim of Quantity, the speaker should proffer the most informative sentence that she has grounds for. Notice that if the embedded clause is undecided in Sara's belief state, then she simply has no belief about

it – there literally is no belief to report. Then, if Sara does have a belief and the speaker is reliably informed about her belief state, they should select the declarative complement clause whose propositional content is positively decided in it.⁶ Thus, in order to maximize informativeness the speaker should use a *ke*-complement conveying that p (or its negation) is positively decided: (23) is ruled out because it is under-informative.

3.3 Nonveridical context

Our account, based on the principle of maximizing informativeness, can explain why polar complements can be embedded under anti-rogative predicates in non-veridical contexts, whereas an account based on a lexical stipulation cannot. Consider first main polar questions, cf. (8) above repeated in (25):

- (24) Sara fekr mikonad ke dustash raftand?
Sara think-3S that friends-poss went-pl
lit. 'Does Sara think that her friends left?'
- (25) Sara fekr mikonad ke aya dustash raftand?
Sara think-3S that Pol.M friends-poss went-pl
lit. 'Does Sara think whether her friends left?'

By definition, questions are uninformative, hence there is no informativeness competition between (24) and (25). On the other hand, there is a dependence relation between the two questions: the issue conveyed by (24) is whether the embedded proposition p is positively decided, whereas in (25) the issue is whether p is decided at all; thus, positive answer to (25) is a pre-condition for (24) to be felicitously asked. In this sense, (24) depends on (25).

- (26) Is p decided in Sara's belief state?
if yes : Is p positively decided in Sara's belief state?
if no : ?? Is p positively decided in Sara's belief state?

It stands to reason that a question does not block another question that it depends on: then, (24) does not block (25). Vice versa, (25) does not block (24) because the latter embodies an issue that requires more information to be resolved, so that its complete answer leads to a more informed common ground for the discourse participants.

Consider then the case of negation in the matrix clause, cf. (9) above repeated in (28):

- (27) Sara fekr nemikone / bavar nadarad ke dustash raftand.
Sara think not-does/ believe not that friends-poss went-pl
- (28) Sara fekr nemikone / bavar nadarad ke aya dustash raftand.
Sara think not-does/ believe not that Pol.M friends-poss went-pl

⁶We assume with Farkas (2003), Roelofsen and Farkas (2015) that positive decidedness constitutes the unmarked case.

As is well known, negation reverses entailment patterns. Therefore, (28) with a polar complement, entails (27) with a declarative complement: if the complement proposition is undecided in Sara’s belief state, *a fortiori* it is also not positively decided. Given this entailment pattern, (27) does not block (28) under the principle of maximizing informativeness. However, the question arises of why (28) does not block (27), as we would expect in our approach. Our tentative answer is that (27) is allowed under the Neg-raising interpretation. Without going into the long-standing debate on the nature of Neg-Raising, we simply point out that it corresponds to an interpretation in which negation has scope over the embedded clause predicate rather than the matrix one. Thus, (27) has both the entailed interpretation (29a) and the Neg-raising interpretation (29b), whereby p is in fact negatively decided in Sara’s belief state:

- (29) a. $\lambda w. \neg \text{BEL}(S, w) \subseteq p$
b. $\lambda w. \text{BEL}(S, w) \subseteq \neg p \iff \lambda w. \text{BEL}(S, w) \cap p = \emptyset$

Not only is the Neg-raising interpretation (29b) not entailed by (28), it actually contradicts it: if p is undecided in Sara’s belief state, the complementary proposition $\neg p$ cannot be positively decided. This being the case, we predict that (27) is not ruled out by (28) only if it receives the Neg-Raising interpretation. This point remains for further investigation.

Consider finally the case in which the attitude report is the prejacent of an epistemic modal, as in (10) repeated in (31):

- (30) Sara bayad fekr konad ke Ali rafte ast.
Sara probably think do-3S that Pol.M Ali gone is
lit. ‘Sara probably thinks whether Ali has left.’
- (31) Sara bayad fekr konad ke aya Ali rafte ast.
Sara probably think do-3S that Pol.M Ali gone is
lit. ‘Sara probably thinks whether Ali has left.’

Following Kratzer (1991, 2012), epistemic modals are based on a stereotypical ordering of the possible worlds accessible from the evaluation world: such worlds are ranked in terms of their congruence to a stereotypical ideal of normalcy. The truth conditions of *probably* q state, informally, that q is a better possibility than $\neg(q)$ with respect to the stereotypical ordering.

From this perspective (30), with a *ke*-complement, conveys that positive decidedness⁷ is a better possibility than not(positive decidedness), the latter including both negative decidedness worlds and undecidedness worlds. Crucially, (30) does not establish any ordering between the latter two, hence it is true in case undecidedness is a better possibility than negative decidedness; however, in this case (31) is false, since it ranks undecidedness below both positive and negative decidedness. Hence, (30) does not entail (31).

⁷Short for: the proposition that the complement proposition is positively decided in Sara’s doxastic state

In turn, (31) do not establish any ordering between positive and negative decidedness, hence it is true in case negative-decidedness is in fact a better possibility than positive decidedness; this circumstance, however, is not compatible with the truth conditions of (30). Therefore, (31) does not entail (30) either.⁸

4 Rogative predicates

Recall that rogative predicates like *porsidan* 'to ask' differ from anti-rogatives in that the polar particle *aya* can never be omitted in the complement clause, as shown in (4)-(5) repeated here as (32):

- (32) Ali az man porsid ke *(aya) dustash raftand.
 Ali from me asked -3S that Pol.M friends-poss went-pl
 'Ali asked me whether his friends left.'

With rogative predicates, the epistemic state that is relevant to assess the the complement clause is not that of the matrix subject, but rather, that of the goal argument. On the other hand, for the asking to be felicitous, the matrix subject must be ignorant as to whether the unique alternative conveyed by the complement clause is positively or negatively decided in the goal's epistemic state. In general, undecidedness seems to be inherent to the lexical semantics of rogative verbs. For this reason, a declarative *ke*-complement conveying positive decidedness cannot be licensed, and *a fortiori*, there is no competition in terms of relative informativeness.

Given that rogative predicates are compatible with wh-questions, but not with *ke*-clauses, we hypothesize that they select for a non-singleton set of propositions. This, however, implies that *ke-aya*-clauses are ambiguous: in addition to the decidedness import defined in (20) above, repeated in (33), they also denote a polar question: (34) is the function that characterizes the polar set $\{p, \neg p\}$ via a disjunctive identity condition.

$$(33) \quad \llbracket ke - aya - S \rrbracket = \lambda q. q \subseteq p \vee q \cap p = \emptyset \\ \rightarrow \lambda q. q \subseteq p \vee q \subseteq \neg p$$

$$(34) \quad \llbracket ke - aya - S \rrbracket = \lambda q. q = p \vee q = \neg p$$

Far from being a shortcoming, this ambiguity allows us to tackle cross-linguistic variation, i.e. the fact that in languages like English, anti-rogative predicates only license declarative complement clauses. Notice that the polar complement denotation in (33) constitutes a weakening of the the polar question

⁸We take *shayad* to convey that q is at least as good a possibility as $\neg q$. Crucially, in Kratzer's characterization, p being a better possibility than q entails p being as good a possibility as q . A similar reasoning applies: (30) with the modal *shayad* requires that positive decidedness be at least as good a possibility as not(positive decidedness), and is true in case positive decidedness is at least as good a possibility as undecidedness, but both are asymmetrically better possibilities than negative decidedness; in this case, however (31) is false.

denotation in (34), in which the identity relation is substituted by the subset relation, so that (33) characterizes a larger set of propositions / attitude states than the polar set of (34). We can then hypothesize Persian differs from languages like English in that it has a morpheme for the polar question operator that also allows for the weaker interpretation in (33).

5 Possible extensions

5.1 Dubitative predicates

An anonymous NACIL3 reviewer pointed out that dubitative predicates select for *ke-aya*-clauses, even though by definition, with these predicates the proposition conveyed by the embedded clause is not decided in the matrix subject's doxastic state. (In this context the disjunctive coda *ya na* 'or not' is obligatory.)

- (35) Sara shak darad / negaran ast ke aya Ali miayad *(ya na)]
 Sara doubt have-3S/ worried is that Pol.M Ali come-3S or not
 'Sara doubts/ is worried whether Ali comes or not.'

It seems plausible to assume that dubitative predicates carry an inherent negative import (as witnessed by their ability to license polarity items in the complement clause in English and in other languages). Carrying this assumption over to Persian,⁹ the denotation of *shak darad* could be characterized as follows:

$$(36) \llbracket shak - darad \rrbracket^w = \lambda P. \lambda y. \neg P(BEL(y, w))$$

When it combines with a polar *ke-aya*-complement, the result is that the proposition *p* conveyed by the polar complement is undecided in the belief state of the matrix attitude holder:

$$(37) \llbracket shak - darad \rrbracket^w (\llbracket ke - aya - S \rrbracket) = \\
\lambda P. \lambda y. \neg P(BEL(y, w)) (\lambda q. q \subseteq p \vee q \cap p = \emptyset) = \\
\lambda y. \neg [(BEL(y, w)) \subseteq p \vee BEL(y, w) \cap p = \emptyset] = \\
\lambda y. \neg [(BEL(y, w)) \subseteq p \ \& \ \neg BEL(y, w) \cap p = \emptyset]$$

However, the obligatoriness of the disjunctive coda is not accounted for. Notice that by this line of reasoning, a *ke*-complement is ruled out as less informative than the polar complement, for the same reason discussed above in connection with negated anti-rogative predicates. We hypothesized above that in the latter contexts a *ke*-complement is only allowed under the Neg-raising interpretation (see above the discussion around (29)). Since dubitative predicates are not Neg-raising, a *ke*-complement is straightforwardly unacceptable.

⁹At the present stage of investigation we cannot provide evidence supporting this extension to Persian, hence this hypothesis remains tentative.

5.2 Emotive factive predicates

The opposite selectional requirement is found with emotive factive predicates like *khushhal* 'happy': here only a declarative complement is allowed.

- (38) Sara khushhal ast ke (*aya) Ali rafte ast.
 Sara happy is that Pol.M Ali gone is
 'Sara is happy that Ali has left.'

Following Heim (1992) and much subsequent literature, predicates of this type presuppose that the proposition in the denotation of the complement clause is positively decided both in the epistemic state of the speaker and in the epistemic state of the matrix subject: therefore, a polar complement conveying undecidedness is not compatible with them.¹⁰

5.3 Progressive aspect

An anonymous NACIL3 reviewer pointed out that in main declarative clauses, a polar complement is licensed under a matrix anti-rogative if the latter is in the progressive aspect:

- (39) Sara darad fekr mikonad ke aya Ali khoshhal ast.
 Sara have-3S think do that Pol.M Ali happy is
 'Sara is thinking whether Ali is happy.'

A crucial observation is that the verb in the progressive form is not interpreted as a stative predicate, as is usually the case, but rather as an activity predicate, describing a mental process: the result state of belief follows the conclusion of the process event (cf. White and Rawlins 2016). In the process event, the complement proposition is undecided in Sara's doxastic state (in fact, it is being decided). This leads to the expectation that with the progressive aspect, a polar complement is not just allowed but, in fact, required. At this stage we cannot provide a compositional analysis for this case, and we leave it for future research. We simply note that the observed aspectual shift is *prima facie* compatible with our characterization of polar complements.

5.4 *Aya* in main questions

In our approach to anti-rogative predicates, we proposed that in embedded contexts, *ke-aya* can introduce either a polar complement or a polar question. When *aya* appears in a main polar question, it is natural to assume that it conveys the second meaning:

¹⁰From a compositional viewpoint, the meaning of emotive factives can be decomposed into an epistemic component and an evaluative component (cf. Anand and Hacquard 2012); the epistemic component would be parallel to the characterization of anti-rogatives proposed above.

- (40) Aya Ali ketab ra khand?
 PM. Ali book Obj.M read-PAST-3S?
 'Did Ali read the book?'

However, it must be stressed that in contemporary spoken Persian *aya* in main questions is actually optional, and when it is present, it adds emphasis to the question. Hence, stipulating a simple alternation with a null morpheme is not entirely satisfactory.

As an alternative, we can build on the analysis proposed by Bianchi and Cruschina (2021) for the so-called *what*-particles optionally appearing in main polar questions in Central and Southern Italian dialects. Following Farkas (2022), the authors assume that canonical questions are characterized by a set of default assumptions, crucially including the assumption that the addressee is knowledgeable about the truth of the prejacent; they propose that the optional *what*-particle strengthens this default assumption to a non-cancellable conventional implicature. By strengthening the default assumption, the speaker emphasizes that they definitely expect an answer from the addressee. Note that in the terms of our current approach, such a conventional implicature amounts to the prejacent of the question being decided, positively or negatively, in the addressee's epistemic state. Thus, adapting Bianchi & Cruschina's approach, we tentatively propose that *aya* in main questions conveys decidedness, but projects it on the non-at-issue layer of conventional implicatures.

6 Quantification over alternatives

In this section, we briefly discuss the logical possibility that *ke aya*-complements always denote polar questions, even when they are embedded under anti-rogative predicates.

In order to maintain a uniform selected type, assume that all clauses denote sets of alternatives (classical propositions), following the approach initiated by Hamblin (1973) and its subsequent developments (see Uegaki 2019 for general discussion). From this perspective, a declarative *ke*-complement would denote a singleton set of alternatives $\{p\}$, while the corresponding polar complement would denote the polar set $\{p, \neg p\}$.

We could then define the matrix anti-rogative predicate as existentially quantifying over the set of alternatives P denoted by the complement clause:¹¹

$$(41) \quad \llbracket f_{\text{ekr}} - \text{kardan} \rrbracket^w = \lambda P. \lambda y. \exists q \in P: \text{BEL}(y, w) \subseteq q$$

Consider again example (6), repeated in (42). Taking $\{p\}$ to be the singleton set denoted by the *ke*-complement, the truth conditions come out as in (42b) (equivalent to (19) above):

¹¹As for universal quantification over alternatives, it is clearly a non-starter: in the case of the polar complement, the matrix attitude state cannot be a subset of both the complementary alternatives.

- (42) a. Sara fekr mikonad ke dustash raftand.
 Sara think does that friends- her went-pl
 b. $\llbracket fekr - kardan \rrbracket^w (\llbracket ke - S \rrbracket) = \exists q \in \{p\}: BEL(S, w) \subseteq q$

The interpretation of example (7), repeated here in (43a), comes out as in (43b): existential quantification over the polar set denoted by the *ke aya*-complement yields truth conditions equivalent to the disjunctive import of decidedness:

- (43) a. Sara fekr mikonad ke aya dustash raftand.
 Sara think does that Pol.M friends- her went-pl
 b. $\llbracket fekr - kardan \rrbracket^w (\llbracket ke - aya - S \rrbracket) = \exists q \in \{p, \neg p\}: BEL(S, w) \subseteq q$

Notice that (42b) entails (43b): hence, our account based on maximization of informativeness in veridical contexts can be maintained. The reversal of the entailment pattern under matrix negation also holds, accounting for (9) above. In the case of the matrix polar questions corresponding to (32) and (43), by non-veridicality the entailment relation between their sentence radicals is irrelevant, so that the principle of maximization of informativeness does not apply, and both questions are allowed.

On the other hand, a problem emerges for the analysis of *ke aya*-complements in the scope of an epistemic modal. We adapt for concreteness the analysis of possibility modals proposed by Aloni (2003): the epistemic possibility modal requires that all the alternatives in the denotation of the prejacent are true in some epistemically accessible world. In (44), we notate with $EP(sp_k, w)$ the speaker's epistemic state at w .

$$(44) \quad \llbracket perhaps - \phi \rrbracket^w = \forall p' \in \llbracket \phi \rrbracket : \exists w' \in EP(sp_k, w) : p'(w') = 1$$

When applied to the prejacent with a *ke*-complement, as in (30) above, we obtain the following truth conditions, which are equivalent to the standard analysis of the epistemic possibility modal. (We notate as $\{p\}$ the set of propositions denoted by the *ke*-complement.)

$$(45) \quad \forall p' \in \{\lambda w''. \exists q \in \{p\}: BEL(S, w'') \subseteq q\}: \exists w' \in EP(sp_k, w): p'(w') = 1 \\ \iff \exists w' \in EP(sp_k, w): BEL(S, w') \subseteq p$$

On the other hand, when the epistemic operator applies in (31), with a *ke aya*-complement, we obtain:

$$(46) \quad \forall p' \in \{\lambda w''. \exists q \in \{p, \neg p\}: BEL(S, w'') \subseteq q\}: \exists w' \in EP(sp_k, w): p'(w') = 1 \\ \iff \exists w' \in EP(sp_k, w): \llbracket \lambda w''. \exists q \in \{p, \neg p\}: BEL(S, w'') \subseteq q \rrbracket (w') = 1 \\ \iff \exists w' \in EP(sp_k, w): \exists q \in \{p, \neg p\}: BEL(S, w') \subseteq q$$

Notice that both in (45) and in (46) the polar set of alternatives is in the scope of the embedded existential operator. Thus, it turns out that (45) – corresponding to (30) with a *ke*-complement – entails (46) – corresponding to (31)

with a polar complement: our account based on maximizing informativeness would wrongly predict that (31) is ruled out, and therefore, it cannot be generalized to this case.

Apart from our specific proposal and from the analysis assumed for epistemic modals, there is a more fundamental problem: if anti-rogative predicates in Persian select for a set of alternatives, it is unclear how to prevent them from selecting a *wh*-question, whose denotation would be of the appropriate semantic type.

Moreover, the alternative set account of clausal complements would essentially obliterate the distinction between anti-rogative and responsive predicates in Persian. This implies that the class that an attitude predicate belongs in is not determined by its inherent semantics, but it is subject to cross-linguistic variation. We regard the former as the null hypothesis, and in this respect, we believe that our first account of Persian polar complements is preferable.

7 Concluding remarks

The proposal that we developed in this paper makes *ke aya*-complements embedded under anti-rogatives in Persian less exceptional than they seem to be at first sight. In our account, anti-rogatives always select for a clause that conveys a single proposition; however, by the import of decidedness of polar *ke aya*-complements, the matrix attitude state entails one of the two complementary propositions p and $\neg p$, which are in the denotation of the corresponding polar question.

We have argued that embedding a polar complement under an anti-rogative predicates is ruled out in veridical contexts by the general principle of maximization of informativeness, because the host sentence is entailed by the corresponding sentence with a declarative *ke*-complements, so that the latter qualifies as strictly more informative. This competition, however, is suspended or reversed in non-veridical contexts, which thus allow for polar complements dembedded under anti-rogatives.

We have reduced the overall distributional pattern of Persian *ke-aya*-clauses to the hypothesis that they convey two different meanings: they denote either polar complements (when embedded under anti-rogatives) or polar questions (when embedded under rogative predicates). We have suggested that the two meanings are related in that the former constitutes a weakening of the latter, and can be seen as a generalization of it to additional embedding contexts.

An ancillary hypothesis in our approach is that anti-rogative predicates do not inherently possess quantificational force, i.e., they do not convey quantification over the worlds of the relevant attitude state: the relation of that attitude state to the embedded proposition is conveyed by the complement clause itself. We are not claiming that this hypothesis is valid of clausal complementation in general, but we believe that it is worth pursuing also at the cross-linguistic level.

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